Toy Models of Superposition

A guided reading of Elhage et al. (2022)

AI & Philosophy Reading Group

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Elhage et al. (2022), *Toy Models of Superposition*, . Original paper

The Problems

A neuron may respond to several apparently unrelated concepts: **polysemanticity**.

One proposed explanation is **superposition**: many feature directions share a smaller activation space.

1. What does it mean to represent a feature?
2. Does model actually learn superposition?
3. Trade-off between discarding a feature and giving it a drection.
4. What geometry results when features share space?

The paper demonstrate superposition can occur than inferred from neuron activations.

Review: linear feature representations

$$h = \sum_{i=1}^{n} a_i f_i, \qquad h, f_i \in \mathbb{R}^d.$$

- A **feature** is an abstract property, such as gender.
- f_i is the **representation** of that property in the activation space.
- a_i is its **activation strength** on this input.
- A neuron is one coordinate h_k of the layer activation.

Prior-reading connection: Park, Choe & Veitch (2023; ICML 2024), *The Linear Representation Hypothesis and the Geometry of Large Language Models*.

What counts as a feature?

Imagine a **sufficiently large model** in which every interpretable feature has its *own* neuron.

Feature	Interpretable property	Its direction
1	Horizontal edge	$e_1 = (1, 0, 0, 0)^T$
2	Dog ear	$e_2 = (0, 1, 0, 0)^T$
3	Negation	$e_3 = (0, 0, 1, 0)^T$
4	Royalty	$e_4 = (0, 0, 0, 1)^T$

Each **feature direction** is *one-hot*: exactly one coordinate is 1.

The activation vector $x = (x_1, \ldots, x_n)^T$ records the strengths of all features.

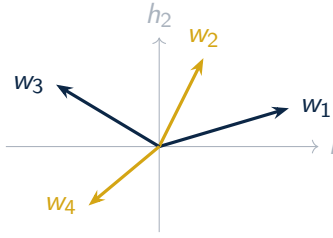
This is an idealized representation.

In a smaller model, features become directions

Now suppose there are n features but only m neurons, with $m < n$.

There is no longer a separate coordinate for every feature.

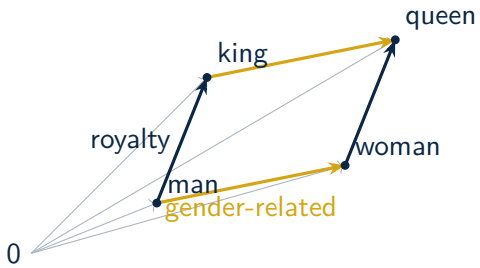
Instead, feature i may be represented by a **direction** $w_i \in \mathbb{R}^m$, which need not line up with a neuron axis.



Four feature directions in only two dimensions.

How feature could be a direction

$$E(\text{woman}) - E(\text{man}) \approx E(\text{queen}) - E(\text{king}).$$



The activation vector is a mixture of features

Let x_i be the activation strength of feature i, and $w_i \in \mathbb{R}^m$ its representation direction.

The activation vector in the smaller model is

$$\boxed{h = \sum_{i=1}^{n} x_i w_i = Wx,}$$

where

$$W = \begin{pmatrix} | & | & & | \\ w_1 & w_2 & \cdots & w_n \\ | & | & & | \end{pmatrix} \in \mathbb{R}^{m \times n}.$$

Changing coordinates without changing the function

Consider two consecutive linear layers:

$$h = Ex, \qquad y = Ah = AEx.$$

For any invertible matrix M, let $E' = ME$ and $A' = AM^{-1}$. Then

$$y' = A'E'x = AM^{-1}MEx = AEx = y.$$

Thus the hidden representation changes, but **the model's input-output function remains exactly the same**.

Individual hidden coordinates need not have intrinsic meanings: we can change them without changing the network's input-output behavior.

ReLU privileges the axes

Let $z = (1, -1)^T$ and let R be a 90° counterclockwise rotation:

$$R = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

$$\mathrm{ReLU}(Rz) = \mathrm{ReLU}\begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix},$$

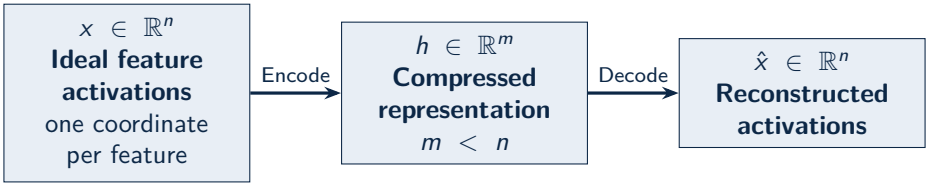
$$R\,\mathrm{ReLU}(z) = R\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

Thus $\mathrm{ReLU}(Rz) \neq R\,\mathrm{ReLU}(z)$: ReLU generally does not commute with rotation.

A privileged basis can make neuron activation meaningful, but it does not prevent superposition.

The experiment: compress, then reconstruct

The experiment models our motivating picture: compress the feature activations of an imagined large model, then try to reconstruct them.



The architecture is

$$\boxed{h = Wx} \qquad \longrightarrow \qquad \boxed{\hat{x} = \mathrm{ReLU}(W^T h + b)}.$$

The network learns $W \in \mathbb{R}^{m \times n}$ and the output bias $b \in \mathbb{R}^n$.

Goal: Recover the original feature activations.

The reconstruction loss

The model is trained to minimize the **weighted squared reconstruction error**:

$$\boxed{L = \sum_{x} \sum_{i=1}^{n} I_i (x_i - \hat{x}_i)^2.}$$

- x_i: original activation of feature i.
- $\hat{x}_i$: reconstructed activation of feature i.
- I_i: importance of feature i.
- $\sum_x$: sum over training inputs.

Features with higher importance receive a larger penalty when reconstructed incorrectly.

Columns of W represent features

Recall that

$$h = Wx, \qquad W = \begin{bmatrix} w_1 & \cdots & w_i & \cdots & w_n \end{bmatrix}.$$

For an input with only feature i activated,

$$x = e_i = (0, \ldots, 0, 1, 0, \ldots, 0)^T.$$

Therefore,

$$\begin{aligned} h = Wx &= \begin{bmatrix} w_1 & \cdots & w_i & \cdots & w_n \end{bmatrix} e_i \\ &= \boxed{w_i}. \end{aligned}$$

Thus, the i-th column of W is precisely the **learned direction** for feature i.

Synthetic features: sparsity and importance

Independently for each feature in the basic experiment,

$$x_i = \begin{cases} 0 & \text{with probability } S_i, \\ U_i, \quad U_i \sim \mathrm{Uniform}[0,1] & \text{with probability } p_i = 1 - S_i. \end{cases}$$

S_i is **sparsity**.

If $S_i = 0.99$, a feature is present on about 1% of inputs. Two independent such features co-occur with probability $0.01^2 = 0.0001$.

Importance $I_i \geq 0$ weights the cost of reconstructing feature i incorrectly.

Why decode with the transpose?

Suppose, for simplicity, that the feature directions $w_1, \ldots, w_n$ are orthogonal and have unit length. Recall that $h = Wx$. Then

$$\begin{aligned} W^T h &= W^T W x \\ &= \begin{bmatrix} w_1 & w_2 & \cdots & w_n \end{bmatrix}^T \begin{bmatrix} w_1 & w_2 & \cdots & w_n \end{bmatrix} x \\ &= I_n x = x. \end{aligned}$$

Thus, in this ideal case, multiplying by W^T perfectly recovers the original feature activations.

But: When $m < n$, the feature directions cannot all be orthogonal. Hence $W^T W \neq I_n$, and reconstruction may introduce interference.

The role of ReLU and bias

When feature directions overlap, other active features may create a small positive signal for a feature that is actually absent.

Recall the reconstruction formula:

$$\hat{x}_i = \text{ReLU}((W^T h)_i + b_i).$$

If $b_i < 0$, the negative bias acts as a **threshold**:

- Small positive signals are pushed below zero and removed by ReLU.
- Stronger signals can still pass through.

Visualize Superposition

Let $G = W^T W$. Since the columns of W are the feature directions $w_1, \ldots, w_n$,

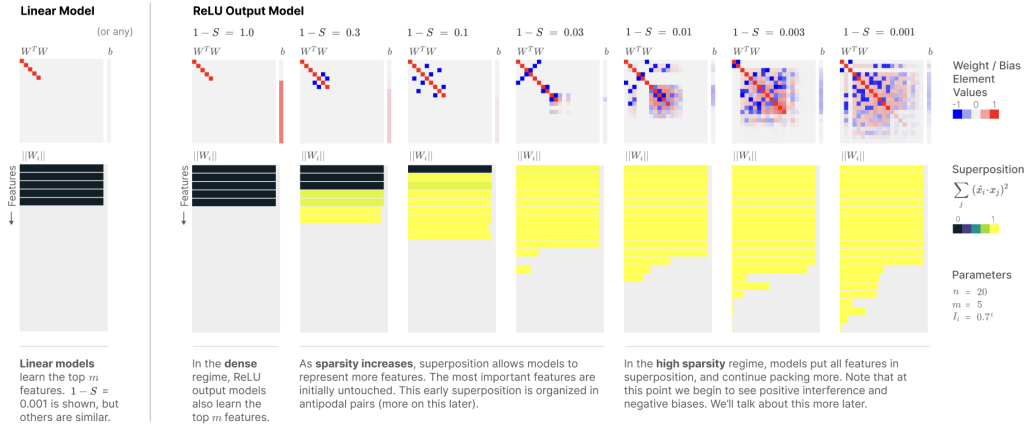
$$\boxed{G_{ij} = w_i \cdot w_j.}$$

Thus, the matrix G tells us two things:

- **Diagonal:** $G_{ii} = \|w_i\|^2$. A value near 1 indicates that feature i is strongly represented; near 0 means it is largely ignored.
- **Off-diagonal:** $G_{ij} = w_i \cdot w_j$. A nonzero value means the two features have overlapping directions.

$W^T W$ allows us to see which features are represented and whether they overlap.

Empirical result: sparsity enables superposition



Every W^TW is 20×20. The colored region grows because more features are represented.

The mathematics: feature loss vs. interference

Consider the simple case where only feature i is activated, with strength 1.

The loss can be written as:

$$L_1 = \underbrace{\sum_i I_i \left(1 - \mathrm{ReLU}(\|w_i\|^2 + b_i)\right)^2}_{\text{Feature loss}} + \underbrace{\sum_{i \neq j} I_j \,\mathrm{ReLU}(w_j \cdot w_i + b_j)^2}_{\text{Interference loss}}.$$

- **Feature loss:** Representing more features helps reduce reconstruction errors.
- **Interference loss:** More represented features create more overlapping directions, potentially causing false activations.

The model balances representing more features against the cost of interference.

Three possible outcomes of a feature

The experiments suggest three possible outcomes for a feature:

1. **Not represented:** The feature is discarded. Its direction is approximately zero.
2. **Superposition:** The feature is represented, but shares dimensions with other features.
3. **Dedicated direction:** The feature has its own direction, orthogonal to the other represented features.

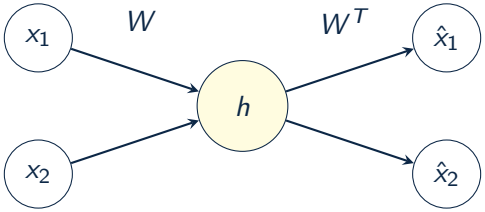
Question: What determines which of these three possibilities the model learns?

The authors discover sharp transitions between these possibilities as sparsity and importance change.

The simplest example: two features, one dimension

$$n = 2, \qquad m = 1.$$

Two input features must be compressed into a single hidden neuron, then reconstructed.



The encoder is now simply

$$W = \begin{bmatrix} w_1 & w_2 \end{bmatrix}, \qquad h = w_1 x_1 + w_2 x_2.$$

The scalar w_i is the representation feature i in the hidden space.

Three possible solutions

1. Represent only feature 1

$$W = \begin{bmatrix} 1 & 0 \end{bmatrix}, \qquad h = x_1.$$

Feature 2 is discarded.

2. Represent only feature 2

$$W = \begin{bmatrix} 0 & 1 \end{bmatrix}, \qquad h = x_2.$$

Feature 1 is discarded.

3. Represent both in superposition

$$W = \begin{bmatrix} 1 & -1 \end{bmatrix}, \qquad h = x_1 - x_2.$$

The two features occupy opposite directions on the same line: an **antipodal pair**.

Why can antipodal superposition work?

Consider $W = \begin{bmatrix} 1 & -1 \end{bmatrix}$, with $\hat{x} = \mathrm{ReLU}(W^T W x)$.

Input x	Hidden state h	Reconstruction $\hat{x}$
$(0,0)$	0	$(0,0)$
$(1,0)$	1	$(1,0)$
$(0,1)$	-1	$(0,1)$
$(1,1)$	0	$(0,0)$

Problem: The model cannot distinguish $(0,0)$ from $(1,1)$.

But: If each feature independently occurs with probability p, then

$$\Pr((1,1)) = p^2.$$

When features are very sparse ($p \ll 1$), simultaneous activation is rare, so superposition can be worthwhile.

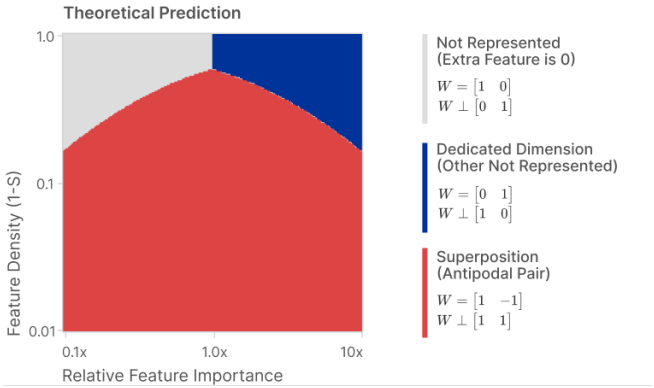
Predicting the phase change

We can now make a prediction.

- **Very sparse features:** They rarely co-occur, so antipodal superposition should work well.
- **Less sparse features:** Simultaneous activation becomes more common, making superposition more costly.
- **Different importances:** When superposition is too costly, the model should retain the more important feature and discard the less important one.

Thus the optimal representation depends on both **sparsity** and **relative feature importance**.

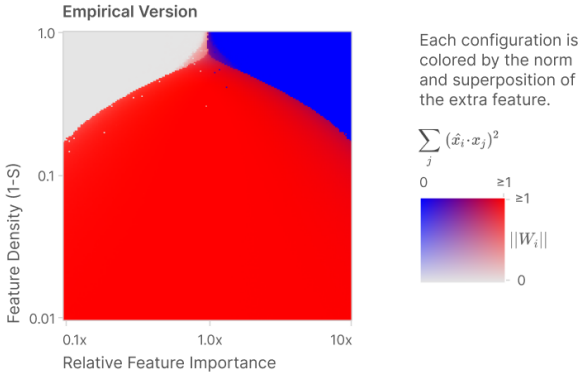
Theoretical prediction: a phase diagram



The theoretical diagram compares the three candidate representations as feature sparsity and relative importance change.

Different regions favor different strategies, with sharp boundaries between them.

Empirical verification: the phase change



The authors train the toy network under different sparsities and relative feature importances.

The empirical phase diagram broadly matches the theoretical prediction.

The transitions between these strategies can be sharp rather than gradual.

The geometry of superposition

We have seen that a model can represent more features than it has dimensions.

Next question: How are these feature directions arranged in the representation space?

The authors investigate two cases:

1. **Uniform superposition:** All features have equal importance and sparsity, and activate independently.
2. **Non-uniform superposition:** Features differ in importance, sparsity, or their correlations.

We begin with the uniform case.

Uniform features: dimensions per feature

Consider $n = 400$ features represented in $m = 30$ hidden dimensions.

How many features does the model actually represent?

Recall that a represented feature usually has $\|w_i\|^2 \approx 1$, whereas a discarded feature has $\|w_i\|^2 \approx 0$.

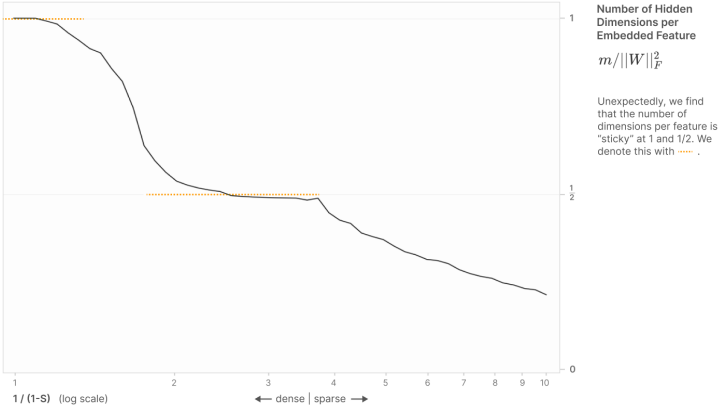
Therefore,

$$\underbrace{\|W\|_F^2 = \sum_i \|w_i\|^2}_{\text{approximately the number of represented features}} .$$

We can estimate the average number of dimensions per represented feature:

$$\boxed{D^* = \frac{m}{\|W\|_F^2}.}$$

An unexpected result: the sticky $1/2$



As features become sparser, the model represents more of them in the same hidden space.

But the curve tends to remain near $D^* = 1$ and $D^* = 1/2$ for extended ranges.

Why $1/2$**?** The model often represents two features as an antipodal pair in one dimension.

How much of a dimension does one feature get?

The previous measure D^* is an average.

We now measure dimensionality for an **individual feature**:

The authors define

$$\boxed{D_i = \frac{\|w_i\|^2}{\sum_j (\hat{w}_i \cdot w_j)^2}, \qquad \hat{w}_i = \frac{w_i}{\|w_i\|}.}$$

Intuition:

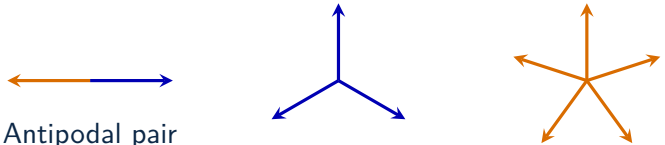
- **Numerator:** How strongly is feature i represented?
- **Denominator:** How much of every feature's representation lies along feature i's direction?

For a dedicated unit direction: $D_i = \frac{1}{1} = 1$.

For two unit antipodal vectors $w_2 = -w_1$: $D_1 = D_2 = \frac{1}{1+1} = \frac{1}{2}$.

Different geometries, different dimensionalities

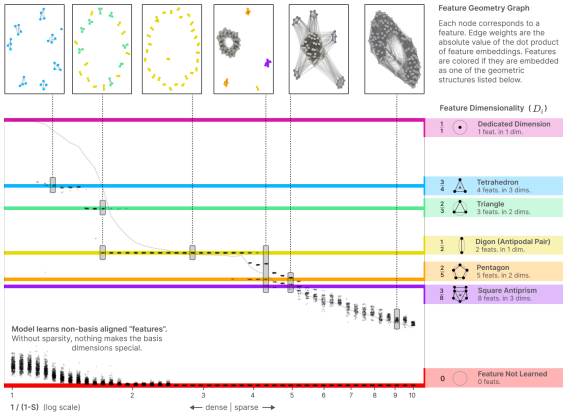
Superposition does not have only one geometry.



For example, in the triangle each unit vector has dot product $-1/2$ with the other two:

$$D_i = \frac{1}{1 + 1/4 + 1/4} = \frac{2}{3}.$$

Empirical result: many geometries of superposition



The individual feature dimensionalities cluster around particular fractions.

These correspond to arrangements such as antipodal pairs, triangles, pentagons, and higher-dimensional polytopes.

What if one feature becomes more important?

So far, all features were equally important and equally sparse.

What happens if we change one feature?

Increasing its importance or density generally increases the pressure to preserve that feature accurately.

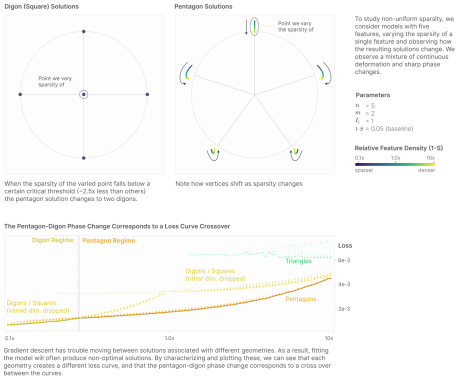
The other feature directions may move away from it, reducing interference.

Decreasing its density makes it less costly to lose or interfere with that feature.

If it becomes sufficiently rare, the model may simply discard it.

The paper demonstrates this by changing the density of one feature in a five-feature, two-dimensional model.

Empirical result: deforming a pentagon



Initially, five features form a pentagon.

- **Denser feature:** Other directions move away from it.
- **Sparser feature:** It gets less space and eventually disappears, leaving two antipodal pairs.

Continuous changes in sparsity can lead to sharp changes in geometry.

Correlated versus anticorrelated features

Another important factor is whether features activate together.

Correlated features: They tend to be present simultaneously.

Suppose two such features have antipodal directions:

$$w_2 = -w_1, \qquad h = x_1 w_1 + x_2 w_2 = (x_1 - x_2) w_1.$$

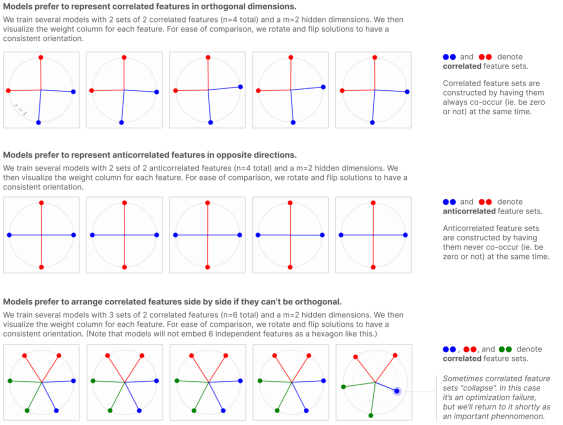
If $x_1 \approx x_2$, the signals nearly cancel!

Thus correlated features generally prefer **orthogonal directions**, if possible.

Anticorrelated features: They rarely or never activate together.

In this case antipodal directions can work well, because the two signals do not cancel.

Empirical geometry: correlations matter



The experiment shows that correlation changes the preferred arrangement of feature directions.

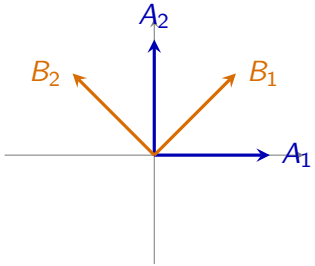
Correlated features tend toward orthogonality; anticorrelated features often share directions with negative interference.

When orthogonality is impossible, correlated features may instead lie closer.

Local almost-orthogonal bases

A particularly interesting finding is that **a globally superposed representation can be locally almost orthogonal**.

Imagine two groups of correlated features:

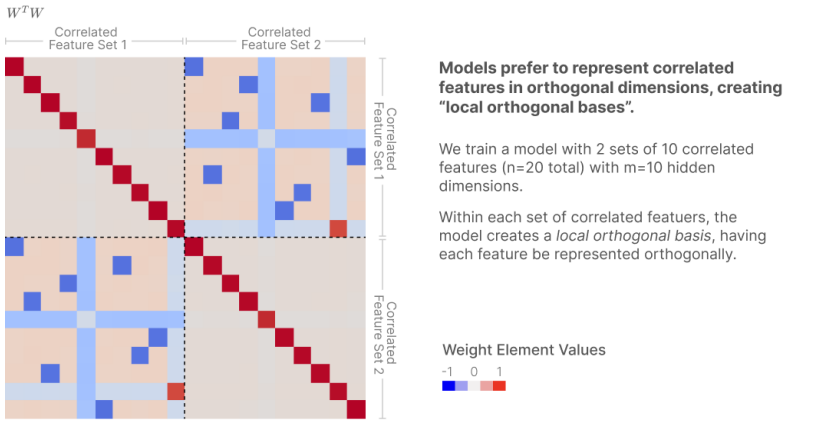


Within each group, the directions are orthogonal:

$$w_{A_1} \cdot w_{A_2} = 0, \qquad w_{B_1} \cdot w_{B_2} = 0.$$

But the two groups overlap with each other.

Empirical evidence: local orthogonal bases



The experiment uses two correlated groups of ten features each, in ten hidden dimensions.

Within each group, the directions are nearly orthogonal.

Across groups, the directions overlap.

Collapsing correlated features

Suppose two correlated features have activations a and b, but the model can preserve only one direction for the pair.

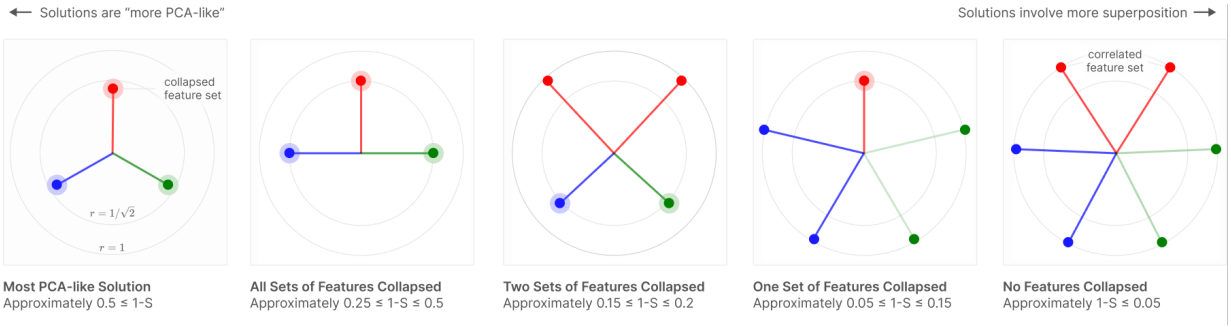
It may represent their **common component**, approximately reconstructing both by their average:

$$(a, b) = (0.8, 0.4) \quad \longrightarrow \quad (\hat{a}, \hat{b}) = (0.6, 0.6).$$

Why does collapsing become preferable?

- **Sparse:** Different correlated pairs rarely activate together, so the model can represent all features separately in superposition.
- **Dense:** Different pairs frequently activate together, increasing interference. Collapsing each pair into one common direction reduces this interference, at the cost of losing differences within each pair.

Empirical result: collapse and PCA



The paper varies the density of six features organized into three correlated pairs.

- **Very sparse:** Six distinct directions form a hexagon.
- **Less sparse:** Correlated pairs progressively collapse toward common directions.
- **Dense:** The solution becomes more PCA-like.

Superposition and PCA-like compression are alternative ways to manage limited capacity.

What the first four sections establish

- Features can become directions in a smaller hidden space.
- Training learns more directions than dimensions when sparsity makes the expected interference cost tolerable.
- Output ReLU filters some interference; biases adjust the sensitivity of the filter.
- Changes in the preferred representation can be sharp.
- Superposition produces regular geometries in uniform models.
- Density, importance, and co-activation influence those geometries; correlated features may collapse into components.

References

Primary reading

Nelson Elhage et al. (2022). *Toy Models of Superposition*. Transformer Circuits.

Original paper and empirical figures arXiv:2209.10652

Previous reading: linear representations

Kiho Park, Yo Joong Choe, and Victor Veitch (2024). *The Linear Representation Hypothesis and the Geometry of Large Language Models*. ICML.

arXiv:2311.03658